

**Department of Computer Science and Engineering**  
**Faculty of Engineering & Technology, Gurukula Kangri Deemed to be University, Haridwar**  
**First Sessional Exam**  
**Subject: Artificial Intelligence**  
**Paper Code: BCE-O534**

**Time: 1 Hr**

**MM: 20**

**Note:** Answer any three questions from Section A (each question carries 4 marks).

Answer any one question from Section B (each question carries 8 marks).

| Q.No. | Section A (Questions of 4 marks each)                                                                                                  | Keyword       | CO    | BT  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|-----|
| Q1.   | Define Artificial Intelligence and explain its main goals.                                                                             | Remembering   | (CO2) | BT1 |
| Q2.   | Describe the key foundations of Artificial Intelligence.                                                                               | Understanding | (CO2) | BT2 |
| Q3.   | What are intelligent agents? Explain their role in AI systems.                                                                         | Understanding | (CO2) | BT2 |
| Q4.   | List and describe the various applications of Artificial Intelligence in real life.                                                    | Applying      | (CO2) | BT3 |
|       | <b>Section B (Questions of 8 marks each)</b>                                                                                           |               |       |     |
| Q1.   | Examine the applications of Natural Language Processing (NLP) in modern AI systems. Provide examples of its use in various industries. | Applying      | (CO3) | BT3 |
| Q2.   | How do AI-based systems process visual data? Discuss the techniques involved in computer vision and its real-world applications.       | Evaluating    | (CO4) | BT5 |

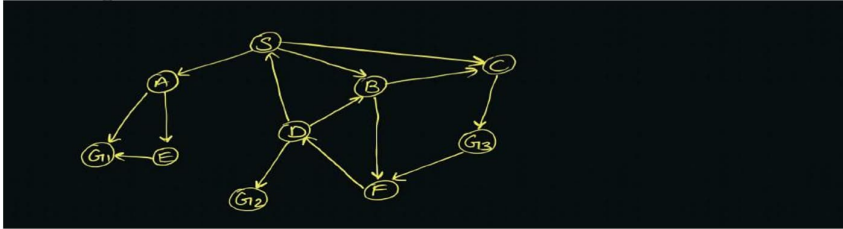
**Department of Computer Science and Engineering**  
**Faculty of Engineering & Technology, Gurukula Kangri Deemed to be University, Haridwar**  
**Second Sessional Exam**  
**Subject: Artificial Intelligence**  
**Paper Code: BCE-O534**

**Time: 1 Hr**

**MM: 20**

**Note:** Answer any three questions from Section A (each question carries 4 marks).

Answer any one question from Section B (each question carries 8 marks).

| Q.No. | Section A (Questions of 4 marks each)                                                                                                                                                                                                                                                                                                                                                                                                  | Keyword       | CO    | BT  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|-----|
| Q1.   | <p>a) A 3-foot tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two stackable, movable, climbable 3-foot high crates.</p>  <p>Action=?, Goals=?</p> <p>b) Describe a state space in which iterative deepening search performs much worse than</p>               | Applying      | (CO3) | BT3 |
| Q2.   | Consider the vacuum world problem. In this problem, instead of two rooms, Agent can move LEFT, RIGHT or suck the trash. Find the number of State, Action, Goal Test, Path Cost and also Create State-Space Graph.                                                                                                                                                                                                                      | Evaluating    | (CO4) | BT5 |
| Q3.   | <p>Consider this state space graph with S as the start node and G1, G2 and G3 are the goal states</p> <p>a) what is this order in which the nodes are removed from the fringe.</p> <p>b) which goal state is reached first.</p>                                                                                                                    | Understanding | (CO2) | BT2 |
| Q4.   | Explain the Bidirectional Search with pseudocode.                                                                                                                                                                                                                                                                                                                                                                                      | Applying      | (CO2) | BT3 |
|       | <b>Section B (Questions of 8 marks each)</b>                                                                                                                                                                                                                                                                                                                                                                                           |               |       |     |
| Q1.   | You have three jugs with capacity 8L, 5L and 3L (litre). 8L jug filled with water and 5L and 3L are empty. Need to measure 4 litre of water. Find the number of State, Action, Goal Test, Path Cost and also Create State-Space Graph.                                                                                                                                                                                                 | Applying      | (CO3) | BT3 |
| Q2.   | <p>Consider a state space graph where the start is number 1 and each state k has two successors: 2K and 2K + 1</p> <p>i) Draw the portion from state 1 to 15</p> <p>ii) Depth limited (l=2, l is equal to 3)</p> <p>iii) iterative deepening.</p> <p>c) Call the action of going from state k to 2K as left and k to 2K + 1 as right. Can you find an algorithm that outputs the solution/path to this problem without any search?</p> | Evaluating    | (CO4) | BT5 |

**Department of Computer Science and Engineering**  
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**Third Sessional Exam**  
**Subject: AI**  
**Paper Code: BCE-O534**

**Time: 1 Hr**

**MM: 20**

**Note:** Answer any two questions from Section A (each question carries 5 marks).

Answer any one question from Section B (each question carries 10 marks).

| Q.No. | Section A (Questions of 5 marks each)                                                                                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1.   | <p>Are they logically equivalent ?</p> <p>a) <math>(p \rightarrow q) \rightarrow r</math> and <math>p \rightarrow (q \rightarrow r)</math></p> <p>b) <math>(p \wedge q) \rightarrow r</math> and <math>(p \rightarrow r) \wedge (q \rightarrow r)</math></p>                                                                    |
| Q2.   | <p><math>p</math> : You have the flu.<br/> <math>q</math> : You miss the final examination.<br/> <math>r</math> : You pass the course.</p> <p>a) <math>\underline{(p \rightarrow \neg r)} \vee \underline{(q \rightarrow \neg r)}</math>,<br/> b) <math>\underline{(p \wedge q)} \vee \underline{(\neg q \wedge r)}</math>.</p> |
| Q3.   | <p>Write the negation of these statements :-</p> <p>a) <math>\forall x (x^2 &gt; x)</math><br/> b) <math>\exists x (x^2 = 2)</math></p>                                                                                                                                                                                         |
|       | Section B (Questions of 10 marks each)                                                                                                                                                                                                                                                                                          |
| Q1.   | <p>Check the consistency of the following statements</p> <p>A : If it is raining, then the ground is wet.<br/> B : It is raining or it is snowing.<br/> C : The ground is not wet.</p> <p>A : If it is sunny, then it is warm.<br/> B : It is sunny or it is snowing.<br/> C : It is not warm.</p>                              |
| Q2.   | <p>Compare and contrast LISP and PROLOG as programming languages for AI applications.</p> <ul style="list-style-type: none"> <li>Which language is better suited for symbolic reasoning, and why?</li> </ul>                                                                                                                    |